technology in modern education, business, and industry.

Exercise

[A] Multiple Choice Questions:

1. Cloud computing delivers computing resources through _____.
   (a) Telephone
   (b) Internet
   (c) Television
   (d) Satellite
2. Which organization provided the standard definition of cloud computing?
   (a) IEEE
   (b) ISO
   (c) NIST
   (d) Microsoft
3. Which company launched Amazon Web Services (AWS)?
   (a) Google
   (b) Microsoft
   (c) IBM
   (d) Amazon
4. Which technology made modern cloud computing possible?
   (a) Compiler
   (b) Virtualization
   (c) Assembler
   (d) Multiprogramming

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5. Cloud computing follows which pricing model?
   (a) Fixed yearly payment
   (b) Pay-as-you-go
   (c) Lifetime payment
   (d) One-time payment
6. Which is NOT a characteristic of cloud computing?
   (a) Resource Pooling
   (b) Rapid Elasticity
   (c) Manual Scaling Only
   (d) On-demand Self-service
7. Which characteristic allows cloud resources to increase or decrease quickly?
   (a) Resource Pooling
   (b) Elasticity
   (c) Security
   (d) Reliability
8. Which cloud characteristic enables multiple users to share the same infrastructure?
   (a) Multi-tenancy
   (b) Backup
   (c) Virtual Memory
   (d) Multiprocessing
9. Cloud computing is mainly accessed through _____.
   (a) USB Drive
   (b) Internet
   (c) CD-ROM
   (d) Bluetooth
10. Which cloud service model provides virtual machines?
   (a) SaaS
   (b) PaaS
   (c) IaaS
   (d) DBaaS
11. Which service model is mainly used by software developers?
   (a) SaaS
   (b) PaaS
   (c) IaaS
   (d) Storage as a Service
12. Google Workspace is an example of _____.
   (a) IaaS
   (b) SaaS
   (c) PaaS
   (d) Virtualization
13. Which is an example of cloud storage?
   (a) Google Drive
   (b) MS Paint
   (c) Notepad
   (d) VLC Player
14. Which cloud architecture component interacts directly with users?
   (a) Back End
   (b) Database
   (c) Front End
   (d) Hypervisor
15. Which layer contains servers and storage?
   (a) Front End
   (b) Network
   (c) Back End
   (d) Client Layer
16. Which is an advantage of cloud computing?
   (a) High hardware cost
   (b) Limited scalability
   (c) Remote accessibility
   (d) Difficult maintenance

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17. Which is a challenge of cloud computing?
(a) High availability
(b) Internet dependency
(c) Resource pooling
(d) Elasticity

18. Which technology reduces latency by processing data near users?
(a) Grid Computing
(b) Edge Computing
(c) Parallel Computing
(d) Cluster Computing

19. Which cloud provider offers EC2?
(a) Microsoft Azure
(b) Google Cloud
(c) AWS
(d) Oracle Cloud

20. Which cloud feature measures resource usage for billing?
(a) Broad Network Access
(b) Measured Service
(c) Automation
(d) Sustainability

Answers

1. (b)	2. (c)	3. (d)	4. (b)	5. (b)
6. (c)	7. (b)	8. (a)	9. (b)	10. (c)
11. (b)	12. (b)	13. (a)	14. (c)	15. (c)
16. (c)	17. (b)	18. (b)	19. (c)	20. (b)

(B) Answer the following questions in short:

1. Define cloud computing.
2. State the NIST definition of cloud computing.
3. List any four characteristics of cloud computing.
4. What is virtualization?
5. Differentiate between cloud computing and traditional IT infrastructure.
6. Write any four advantages of cloud computing.
7. Explain the front end and back end of cloud architecture.
8. What are the three cloud service models?
9. List any four challenges of cloud computing.
10. Give any four real-world applications of cloud computing.

(C) Answer the following questions in detail:

1. Explain the concept of cloud computing. Discuss its evolution from the mainframe era to cloud-native computing.
2. Explain the characteristics of cloud computing with suitable examples.
3. Compare cloud computing and traditional IT infrastructure in detail.
4. Explain the advantages and benefits of cloud computing.

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5. Discuss the major challenges of cloud computing and suggest possible solutions.
6. Explain the architecture of cloud computing with a neat diagram and describe the functions of the front end, back end, and network.
7. Explain the three cloud service models (IaaS, PaaS, and SaaS) with characteristics, benefits, and examples.
8. Describe the evolution of cloud computing, highlighting the four major eras of development.
9. Explain cloud storage and online collaboration tools with suitable real-world applications and examples.
10. Discuss the role of cloud computing in supporting emerging technologies.



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